Physics and Mathematics - Symbol-Rich MCQ Set

Greek letters, units, vectors, equations and image-based formula options

1. A wave has frequency $f = 50$ Hz and wavelength $\lambda = 2$ m. What is its speed?

A. 25 m/s B. 52 m/s C. 100 m/s D. 2500 m/s

(2) Which equation represents mass-energy equivalence?

$$E = mc^2$$

(a) $F = ma$ (b) $p = mv$ (c) $E = mc^2$ (d) $V = IR$

Q3. The displacement of an object is given by the following relation.
Select the correct formula.

The object starts with initial velocity u and acceleration a.

$$v = u + at$$

A: velocity equation (image-based option)

$$s = ut + \frac{1}{2}at^2$$

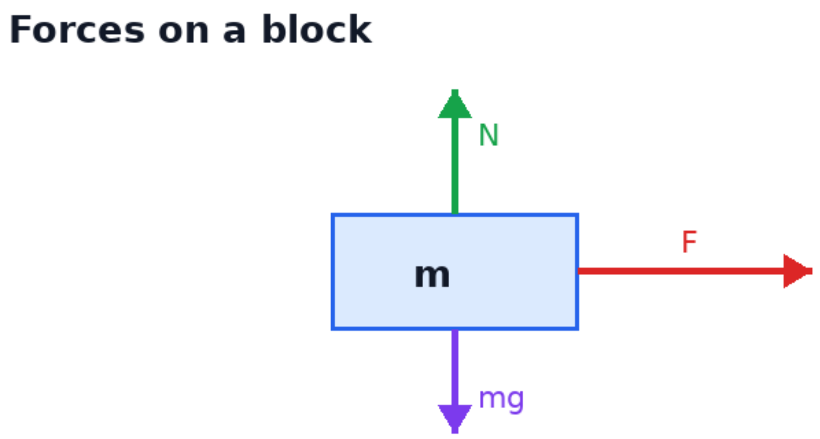
B: displacement equation (image-based option)

4) Which quantity is a vector?

Temperature, Mass, Force, Time

Unlabelled inline options, separated by commas.

Question 5. Refer to the diagram. Which force acts vertically downward on the block?



i. Applied force F

ii. Normal force N

iii. Weight mg

iv. No force acts downward

6. Evaluate: $\int_0^2 x \, dx$

1, 2, 3, 4

Numeric options are comma-separated without labels.

7) Which symbol denotes angular frequency?

ω α θ Δ

Greek-symbol-only options in a horizontal layout.